EMBEDDED SYSTEMS AND ELECTRONICS ENGINEER

2 YEARS OF EXPERIENCE

Julien G.

- DESIGN AND INTEGRATION OF EMBEDDED SYSTEMS INVOLVING SENSORS AND DATA ACQUISITION CHAINS. LOW LEVEL SOFTWARE DEVELOPMENT IN C ON STM32 PLATFORMS. IMPLEMENTATION OF COMMUNICATION INTERFACES SUCH AS I2C AND SPI. PROTOTYPING AND VALIDATION OF REAL TIME FUNCTIONS. FPGA DEVELOPMENT IN VHDL FOR EMBEDDED SIGNAL PROCESSING. TEST AUTOMATION AND TECHNICAL WORKFLOWS USING PYTHON, BASH, AND GIT. ELECTRICAL FAULT DIAGNOSIS, FUNCTIONAL VALIDATION, AND SYSTEM RELIABILITY IMPROVEMENT. ABILITY TO FORMALIZE TECHNICAL NEEDS, STRUCTURE DOCUMENTATION, AND CONNECT FIELD CONSTRAINTS WITH ELECTRONICS, EMBEDDED SOFTWARE, AND OVERALL SYSTEM PERFORMANCE.
- TECHNICAL RIGOR, ANALYTICAL THINKING, AUTONOMY, ADAPTABILITY, CURIOSITY, DIAGNOSTIC MINDSET, CLEAR COMMUNICATION, CROSS FUNCTIONAL COLLABORATION, EXECUTION RELIABILITY, AND FAST LEARNING IN NEW TECHNICAL ENVIRONMENTS.

Engineer focused on embedded systems, electronics, and real time processing, with experience in sensor integration, system validation, FPGA development, and low level software. Contributes to projects involving data acquisition, hardware interfaces, reliability improvement, test automation, and technical documentation. Looking for assignments combining upstream technical definition, integration, validation, and development of embedded or electronic systems.

BUSINESS AREAS

Embedded System	Analog Electronics / Application Software / Automation (control command) / Embedded Software / System (architecture, modeling, simulation) / Digital Electronics / Power Electronics / Radio/Frequency / Microwave / Communication Systems / HW / SW Test Benches
Electrical Engineering	Electrical Engineering (Design / Sizing) / Optronics / Optics / Signal Processing
Others	Embedded Systems / Electrical Engineering / Signal Processing / Applied AI for embedded systems

SKILLS

Digital analog electronics	Oscilloscope / MOSFET / PSPICE / ALTIUM / System verilog

Skills file

Tools / Softwares / ERP	GITHUB / Git
Bus / System / Network / ToIP	SPI / I2C
Programmable components	FPGA / Xilinx
Target platform	ESP32 / Raspberry / ATmega32
Languages	Automation-scripting / C++ / Python / Matlab / Bash / VHDL
CAD	CADENCE / Kicad
Real time	DSP / Bootloader / Embedded firmware / Real time data acquisition / Control loop / Sensor fusion / Real time processing / Latency constraints / RTOS
Operating Systems	iOS
Design	Embedded systems architecture / Sensor integration / Data acquisition chain / Functional validation / Electrical fault diagnosis / Signal processing / System reliability / Technical documentation
Framework / API / Webservices / CMS	API / Webservices / CMS TensorFlow / PyTorch
Others	Sensors / IMU / MPU6050 / Model quantization / Edge AI / CNN / Embedded Linux / Automation scripts / Electrical integration / IVVQ

EDUCATION

2029 **Engineering Program S3E (Master's level)**

2026 - 2029 / CESI Nanterre

2025 **Bachelor in Electrical Engineering**

2022 - 2025 / Johns Hopkins University

LANGUAGES

French

Native speaker

English

C2: experienced - mastery

CERTIFICATIONS & TECHNICAL TESTS

2022 **Driver's License B**

PROFESSIONAL EXPERIENCE

Cable Assembler

Period	January 2026 - Until now (6 months)
Client	ELECTRO FAISCEAUX
Project	**Cable Assembler**
Description	Assembly and integration of complex electrical harnesses for industrial systems and critical medical devices. Diagnosis, correction, and validation of electrical faults in a regulated environment with strong quality, traceability, and compliance requirements.
Teams & methods	Worked in coordination with production, integration, and quality teams, following technical instructions, assembly procedures, fault analysis steps, and validation requirements in a regulated industrial environment.
Tasks	Assembled and integrated complex electrical harnesses for industrial and medical systems. Identified, corrected, and validated electrical faults. Followed quality and traceability procedures. Contributed to compliance and reliability checks before final validation.
Tools & technologies	Electrical harness integration, cabling, technical drawings and work instructions, electrical fault diagnosis, quality control, traceability, compliance, regulated industrial environment, critical medical devices.

Laboratory Assistant

Period	January 2024 - December 2025 (2 years)
Client	JOHNS HOPKINS UNIVERSITY
Project	**Laboratory Assistant**
Description	Designed and executed validation protocols for precision electronic systems. Structured and formalized technical documentation to support experimental reproducibility.
Teams & methods	Worked in a laboratory environment alongside researchers, instructors, and technical staff. Followed a rigorous validation approach based on test preparation, protocol execution, result analysis, traceability of observations, and structured documentation.
Tasks	Prepared and executed validation tests on precision electronic systems. Recorded and analyzed experimental results. Ensured traceability of observations and test conditions. Produced structured technical documentation to support repeatability and reproducibility.

Tools & technologies	Experimental validation / Test protocols / Laboratory instrumentation / Precision electronic systems / Technical documentation / Result analysis / Traceability / Experimental reproducibility / Oscilloscope

Research Intern

Period	January 2024 - December 2025 (2 years)
Client	RISE@APL (JOHNS HOPKINS APL)
Project	**Research Intern**
Description	Contributed to an applied research project focused on MRI data exploitation for the representation and analysis of the brain connectome. The work involved processing complex neuroimaging data, structuring graph based representations, and analyzing relationships between brain regions. The objective was to help build tools that transform MRI derived data into usable representations for connectivity analysis, with attention to pipeline robustness and graph interpretation.
Teams & methods	Worked within a multidisciplinary research team involving scientific and technical contributors focused on imaging, data analysis, and computational modeling. Used an iterative approach based on data preparation, scripting, graph analysis, result checking, and reproducible workflows.
Tasks	Prepared and structured MRI derived data for analysis pipelines. Contributed to modeling the connectome as a graph representing regions and their connections. Used graph analysis tools to explore network structure and extract topological properties. Developed scripts to automate parts of the workflow. Checked input data consistency and result quality. Contributed to reproducible code and technical documentation.
Tools & technologies	Python, NetworkX, graph analysis, MRI data, connectome mapping, data processing, network visualization, scientific scripting, pipeline automation, connectivity analysis

Researcher

Period	January 2021 - December 2023 (3 years)
Client	CSMS LAB (JOHNS HOPKINS)
Project	**Researcher**
Description	Research work on local learning algorithms as scalable alternatives to backpropagation, with a focus on high dimensional systems and physically inspired computation. Explored links between learning theory

		and system level constraints in neuromorphic, distributed, and sensor based settings.
	Teams & methods	Worked within a multidisciplinary research environment involving faculty, researchers, and technical contributors focused on learning systems, sensing, and computational modeling. Followed an iterative research method combining literature review, algorithm prototyping, experimental evaluation, and technical documentation
	Tasks	Studied and developed local learning approaches as alternatives to backpropagation. Investigated properties of high dimensional systems. Explored connections between theoretical models and physically inspired computing systems. Structured technical work and results in a reproducible research workflow.
	Tools & technologies	Machine learning, local learning algorithms, high dimensional systems, computational modeling, distributed computation, neuromorphic inspired systems, intelligent sensing, experimental research workflows, technical documentation

PERSONAL/ACADEMIC PROJECTS

Embedded Linux Oximeter Pipeline

Description	Design of a complete embedded acquisition and processing chain for physiological signals used in SpO2 estimation. The system relied on an SPI sensor interface and a software processing chain running in an embedded Linux environment. The objective was to build a reliable, testable, and reproducible pipeline from raw acquisition to final measurement.
Tasks	Designed the sensor data acquisition chain. Integrated SPI communication with the measurement component. Developed filtering and signal processing steps required for SpO2 estimation. Structured the software pipeline on embedded Linux. Automated tests and processing using Python and Bash scripts. Used Git for version control and project tracking. Verified the functional and technical behavior of the full chain.
Tools & technologies	Embedded Linux, SPI, Python, Bash, Git, signal filtering, SpO2 computation, sensor data acquisition

Drone Sensor Fusion and PID Control

Description	Development of an embedded system for a drone integrating sensor acquisition, sensor fusion, and real time control. The objective was to ensure stable platform behavior using an IMU connected to an STM32 microcontroller under strict timing constraints.
Tasks	Developed embedded firmware in C on STM32. Integrated an MPU6050 inertial sensor through I2C. Built the sensor acquisition and processing chain. Implemented sensor fusion logic to improve state estimation. Implemented and tuned a PID control loop for stabilization. Validated real time behavior with a loop target below 10 ms. Performed testing and tuning to improve stability and robustness.
Tools & technologies	STM32, C, MPU6050, I2C, IMU, sensor fusion, PID, embedded firmware, real time control

Wavelet Processing on FPGA

Description	Development of an FPGA implementation of a discrete wavelet transform for real time image processing, with a focus on infrared image denoising. The objective was to design a pipelined hardware architecture meeting throughput and robustness constraints while demonstrating the value of wavelet based preprocessing.

Tasks	Designed the processing architecture in VHDL on FPGA. Implemented a pipelined chain for real time execution of the transform. Defined and tested the hardware logic associated with signal processing. Performed functional validation on an infrared imaging use case. Analyzed denoising quality and overall chain performance. Produced technical documentation for the implementation choices.
Tools & technologies	FPGA, VHDL, DWT, signal processing, pipeline architecture, real time processing, infrared image denoising

CNN for Image Classification

Description	Design and optimization of a convolutional neural network pipeline for image classification. The work focused on training, evaluation, and adaptation of the model to memory and computation constraints closer to embedded deployment. The objective was to reach a useful compromise between accuracy, inference latency, and memory footprint on datasets such as CIFAR 10 and MNIST.
Tasks	Designed and trained CNN architectures for image classification. Prepared data and built the experimentation pipeline. Evaluated model performance using accuracy and computational cost metrics. Optimized the architecture to reduce latency and memory usage. Worked on model quantization and adaptation to embedded constraints. Used Python and deep learning frameworks for prototyping and testing.
Tools & technologies	Python, TensorFlow, PyTorch, CNN, model quantization, image classification, embedded optimization